Computer Science

	Normative 4-year Degree
Major	Computer Science 計算機科學
Degree / Award Title	Bachelor of Science in Computer Science 理學士(計算機科學)
Offering Academic Unit	Department of Computer Science
Normal Period of Study	4 years
Maximum Period of Study	8 years
Credit Units Required for Graduation	122 credit units

Note: The following curriculum information is subject to periodic review and changes.

GE Requirements (31 credit units) Catalogue Term : Semester A 2025/26

		Credit Units
GE University Requirements	GE1401 University English	3
	GE2410 English for Engineering	3
	GE1501 Chinese Civilisation - History and Philosophy	3
	GE1601 Whole-Person Development	1
GE Distributional Requirements	Take at least 1 course from each of the three distributional areas: Area 1: Arts and Humanities Area 2: Study of Societies, Social and Business Organisations Area 3: Science and Technology Visit the GE website for more course information.	12

College-specified Courses	CS1302 Introduction to Computer Programming	3
	CS3402 Database Systems	3
	JC2066 IT Professionals: Ethical, Legal and Social Issues	3

College / School Requirements (8 credit units) Catalogue Term : Semester A 2025/26

Course Code	Course Title	Credit Units
MA1503	Linear Algebra with Applications	4
MA1508	Calculus	4

Major Requirements (83 credit units) Catalogue Term : Semester A 2025/26

1. Core Courses (59 credit units)

Required CS Courses - 50 credit units

Course Code	Course Title	Credit Units	Remarks
CS2115	Computer Organization	3	
CS2204	Fundamentals of Internet Applications Development	3	
CS2310	Computer Programming	3	
CS2312	Problem Solving and Programming	3	
CS2402	Introduction to Computational Probability Modeling	3	
CS2613	Seminars on Contemporary Technology	2	
CS3103	Operating Systems	3	
CS3201	Computer Networks	3	
CS3334	Data Structures	3	
CS3342	Software Design	3	
CS4335	Design and Analysis of Algorithms	3	
CS4514	Project	9	A course to fulfil the internship/consultancy project/research project requirement for normative 4-year degree students

Choose either CS3000 or CS3505:

Course Code	Course Title	Credit Units	Remarks
CS3000	Professional Career Development Internship	0	Exclusive with CS3505 Students who choose to enrol in CS3000 are required to take 3 elective courses from the

			Elective List in lieu of the 9-credit CS3505.
CS3505	IT Professional Internship	9	Exclusive with CS3000 A course to fulfil the internship/consultancy project/research project requirement for normative 4-year degree students @Same course

Required Supporting Courses – 9 credit units

Course Code	Course Title	Credit Units	Remarks
MA2185	Discrete Mathematics	3	

Choose any ONE from the following list:

Course Code	Course Title	Credit Units	Remarks
GE2313	Global IT Case Studies	3	
GE2324	The Art and Science of Data	3	
GE2338	Everyday Security - Protecting Yourself in the Digital Age	3	
GE2340	Artificial Intelligence – Past, Present, and Future	3	

Choose any ONE from the following list:

Course Code	Course Title	Credit Units	Remarks
CB2100	Introduction to Financial Accounting	3	
CB2300	Management	3	
CB2500	Information Management	3	
CB2601	Marketing	3	

2. Electives (24 credit units)

Electives: *minimum **24** credit units from these electives*
Students may choose any of the streams by taking **3** courses of the selected stream and any **5** elective courses from the list. For those who do not want to focus on a selected stream, they can take any **8** elective courses from the list.

Artificial Intelligence Stream : Stream Core

Course Code	Course Title	Credit Units	Remarks
CS4486	Artificial Intelligence	3	
CS4487	Machine Learning	3	*Same course

Choose one out of the following six courses:

Course Code	Course Title	Credit Units	Remarks
CS3383	AI for Software Engineering	3	New course
CS4186	Computer Vision and Image Processing	3	#Same course
CS4386	AI Game Programming	3	
CS4489	Optimization Foundations for AI Training	3	New course
CS4490	Generative AI Essentials and Applications	3	New course
CS4493	Natural Language Processing	3	

Data Science Stream: Stream Core

Course Code	Course Title	Credit Units	Remarks
CS3481	Fundamentals of Data Science	3	
CS4480	Data-Intensive Computing	3	
CS4487	Machine Learning	3	*Same course

Cybersecurity Stream : Stream Core

Course Code	Course Title	Credit Units	Remarks
CS4286	Internet Security and e-Commerce Protocols	3	
CS4293	Topics in Cybersecurity	3	
CS4394	Information Security and Management	3	

Multimedia Computing Stream : Stream Core

Course Code	Course Title	Credit Units	Remarks
CS3483	Multimodal Interface Design	3	
CS4182	Computer Graphics	3	

Choose one out of the following five courses:

Course Code	Course Title	Credit Units	Remarks
CS3189	User-centred Interaction Design	3	
CS4185	Multimedia Technologies and Applications	3	

CS4186	Computer Vision and Image Processing	3	#Same course
CS4187	Computer Vision for Interactivity	3	
CS4188	Virtual Reality	3	

Software Engineering and Project Management Stream : Stream Core

Choose three out of the following four courses:

Course Code	Course Title	Credit Units	Remarks
CS3346	Software Testing and Maintenance	3	
CS3356	Managing Software Projects	3	Exclusive with IS4500
CS4348	Software Quality Management	3	
CS4389	Decentralized Applications Development	3	

Other Electives :

Course Code	Course Title	Credit Units	Remarks
CS3283	Distributed Systems	3	
CS3382	Web Usability Design and Engineering	3	
CS3391	Advanced Programming	3	
CS4280	Advanced Internet Applications Development	3	
CS4284	Mobile Computing	3	
CS4288	Cryptographic Algorithms and Protocols	3	
CS4295	Mobile Application Programming	3	
CS4296	Cloud Computing	3	
CS4298	iOS Application Development	3	
CS4367	Computer Games Design	3	
CS4381	Advanced Software Design	3	
CS4385	Topics in Software Engineering	3	
CS4485	Information Retrieval	3	
CS4552	Guided Study	3	A course to fulfil the internship/consultancy project/research project requirement for normative 4-year degree students

MA2172	Applied Statistics for Sciences and Engineering	3	

3. AI, Computing and Transformation Stream (only for students admitted via 1070 (JS1070))

AI, Computing and Transformation Stream : Stream Core

Course Code	Course Title	Credit Units	Remarks
CS3505	IT Professional Internship	9	@Same course
SDSC3026	International Professional Development	3	

Suggested Study Plan

Suggested Study Plan for BSc Computer Science